



CARLA CORDOBA
ARGENTINA
2026

LATIN AMERICA HIGH PERFORMANCE COMPUTING CONFERENCE

September 21-25

Recent progress of WMO Integrated Processing and Prediction System (WIPPS)

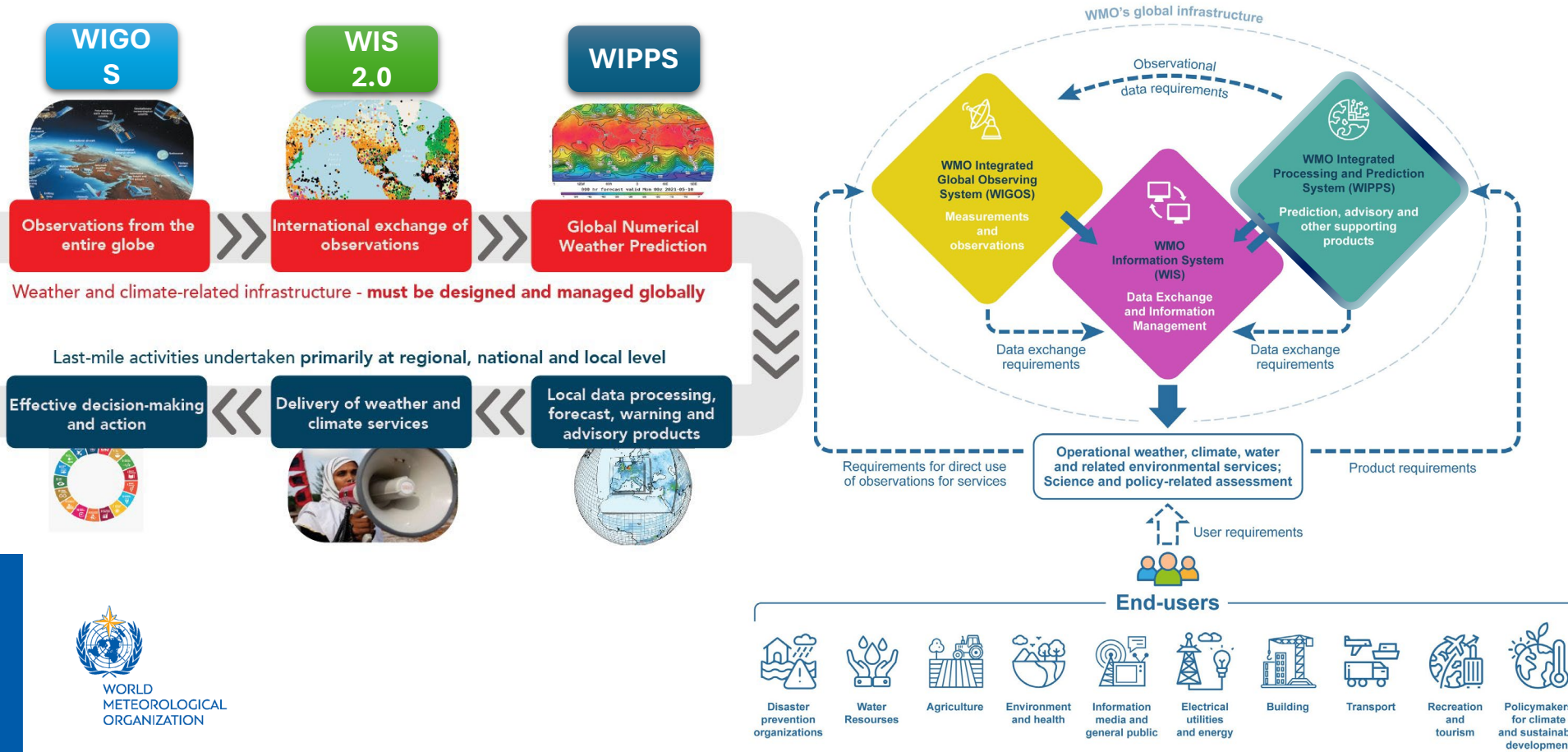
Yuki Honda, Chief of WIPPS, WMO

World Meteorological Organization

- The World Meteorological Organization (WMO) is a specialized agency of the United Nations.
- WMO provides the framework for international cooperation to advance meteorological, climatological, hydrological, and related environmental services, to improve well-being of all.
- The WMO consists of 193 Members including 6 Territories.



WMO Meteorological Value Cycle and WMO Global Infrastructure



WIPPS Activities: Full list

General purpose activities (14)

- Global deterministic numerical weather prediction
- Limited area deterministic numerical weather prediction
- Global ensemble numerical weather prediction
- Limited area ensemble numerical weather prediction
- Global numerical sub-seasonal prediction
- Global numerical seasonal prediction
- Annual to decadal climate prediction
- Global climate reanalysis
- Numerical ocean wave prediction
- Global numerical ocean prediction
- Global numerical storm surge prediction
- Nowcasting
- Sub-seasonal to seasonal hydrological prediction
- Snow cover prediction

Specialized activities (15)

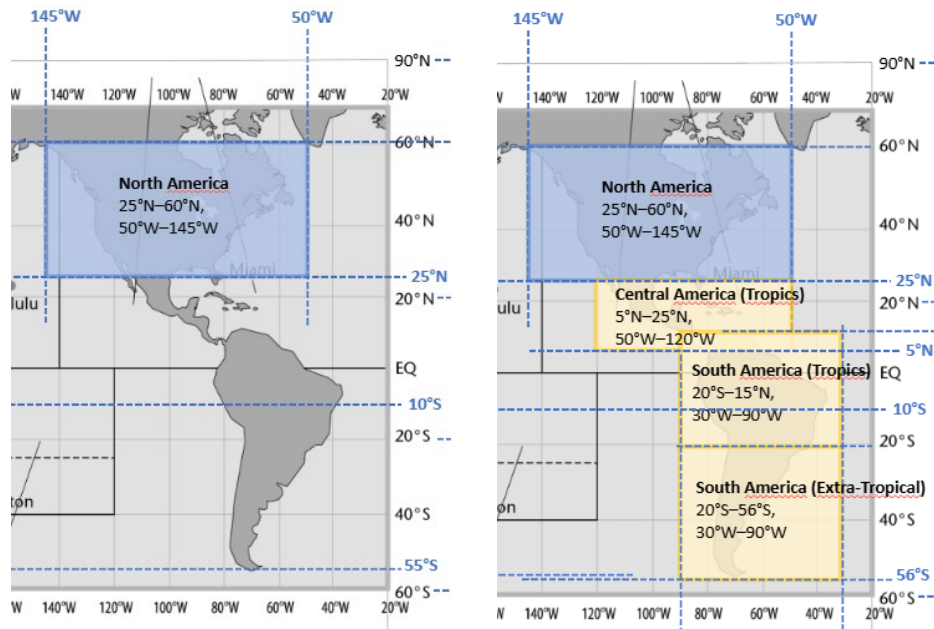
- Regional climate prediction and monitoring (RCC)
- Coordination of multi-model ensemble for sub-seasonal prediction
- Coordination of multi-model ensemble for seasonal prediction
- Coordination of annual to decadal climate prediction
- Coordination of assessment of multiple climate reanalysis
- Regional severe weather forecasting
- Tropical cyclone forecasting, including marine-related hazards
- Nuclear environmental emergency response
- Non-nuclear environmental emergency response
- Atmospheric sand and dust storm forecasts
- Vegetation fire and smoke pollution forecasts
- Volcano watch services for international air navigation
- Marine meteorological services
- Marine environmental emergency response
- Flash flood forecasting

Non-real-time activities (5)

- Coordination of deterministic numerical weather prediction (NWP) verification
- Coordination of ensemble prediction system (EPS) verification
- Coordination of wave forecast verification
- Coordination of tropical cyclone forecast verification
- Coordination of observation monitoring

Standardized verification for NWP outputs

- [**Revised**] Standardized verification (SV) for global deterministic and ensemble NWP.
 - Verification against analyses, upper-air and surface observations.
- [**NEW**] SV for limited-area deterministic and ensemble NWP
 - Verification against upper-air and surface observations.
- The proposed changes need to be approved by INFCOM



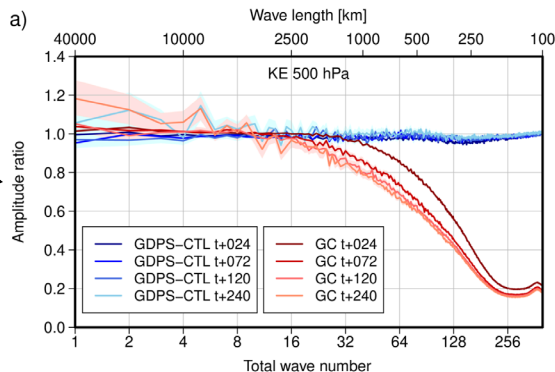
Verification of data-driven (ML) forecasts

- LCs will allow scores from **two forecasts per contributing centre** (e.g. physics-based + ML)
- Both forecasts have to be **fully operational** and products made available
- How to ensure **fair comparison**, as ML models more explicitly minimize some of the scores?

→ Measure **forecast activity**: SDEV of forecast anomaly (currently RMS of forecast anomaly)

→ **Fair CRPS** in addition to CRPS (to account for number of ENS members)

→ Evaluate power spectra / spectral bands: to be included under 'recommended' for DET



Husain et al (2025)

<https://doi.org/10.1175/WAF-D-24-0139.1>

AI Exploration Roadmap for WIPPS

- AI-based Earth System Prediction (AI-ESP) continues to demonstrate improving skills.
- Uncertainties remain regarding the extent to which AI technologies can be incorporated into, or replace, existing applications such as NW in meteorology.

Identifying Key Issues and Challenges

Six foundational pillars for safe AI integration



1. Data Availability & Quality

Securing long-term climate archives and addressing global observational gaps.



2. Prediction Dynamics

Navigating high-resolution localization and the absolute necessity of post-processing.



3. Verification

Rethinking accuracy beyond standard physical laws for rare extremes.



4. Infrastructure

Scaling operational computational requirements and human capacity at local NMHSs.



5. Explainability

Ensuring forecaster trust, model transparency, and operational documentation.



6. Framework Evolution

Adapting WIPPS Technical Regulations and global data exchange protocols.

Issues and Challenges for AI use

by NMHSs and WIPPS-DCs, particularly in the context of **EW4All**

1. Data Availability and Quality

1. Provision of long-term observation/analysis dataset for AI training and verification
2. Observation requirements for high-impact forecasting
3. Quality control of observations

2. Prediction and post-processing

1. Benefit and applicability of AI-ESP for nowcasting and forecasting
2. High-resolution AI-based forecasts for local area
3. Compatibility between initial fields and training dataset
4. High-resolution data-driven models
5. Benefit and necessity of post-processing

3. Verification

1. WIPPS standard verification of AI-ESP model outputs
2. Verification of AI-ESP model outputs for local weather elements and extremes
3. Verification of meteorological consistency between variables of AI-ESP model outputs
4. Verification of forecast scenarios

4. NMHS infrastructure and capacity requirements

1. Use of AI-ESP model outputs in operational forecasting and warning
2. Implementation and maintenance of AI-based systems

5. Model Explainability and Transparency

1. Guidelines on the use and interpretation of AI-ESP model outputs
2. Guidelines on developing explainable AI-ESP

6. WIPPS Framework and Technical Regulations

1. Expansion of WIPPS activities to accommodate AI-ESP model outputs
2. Impact of AI on the WIPPS cascading process

6 categories

18 sub-categories and

52 questions

AI-related WIPPS Pilot Projects and other WMO activities

Testing through WIPPS Pilot Projects <https://community.wmo.int/en/wipps-pilot-project>

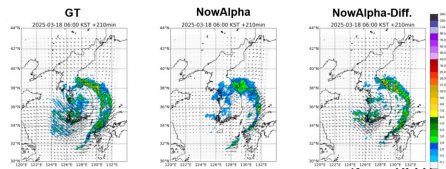
Key issues and challenges identified will be addressed through pilot projects. Each of these pilot projects will serve as a proof of concept for broader AI integration into WIPPS and will be designed to test the scalability and effectiveness of AI solutions in operational settings.

Nowcast

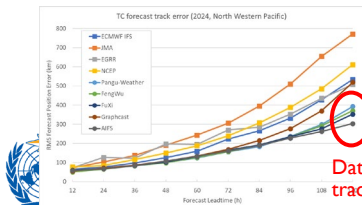
Short to Medium

Sub- to Seasonal

AI for Nowcasting Pilot Project



(from KMA)



Data-driven models tracks generally outperformed traditional NWP.

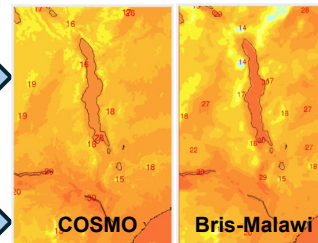
WORLD
METEOROLOGICAL
ORGANIZATION

Data-Driven Weather Forecasting for All (Malawi Project)

AI in TC Analysis and Prediction

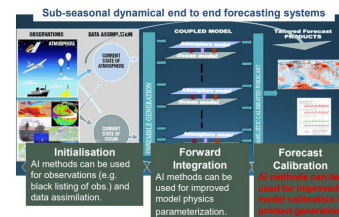
Weather Prediction Model Intercomparison Project

Global Riverine Flood Prediction Pilot



(from MET Norway and DCCMS Malawi)

AI Weather Quest



AI methods as alternatives to dynamical models (from AI Weather Quest)

AINPP in Latin America

Current Activities in AI-Based Nowcasting

Lightning • Precipitation • Verification • Research-to-Operations

Alan J. P. Calheiros — INPE/Brazil

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MINISTÉRIO DA
CIÊNCIA, TECNOLOGIA
E INOVAÇÃO



UNião e Reconstrução

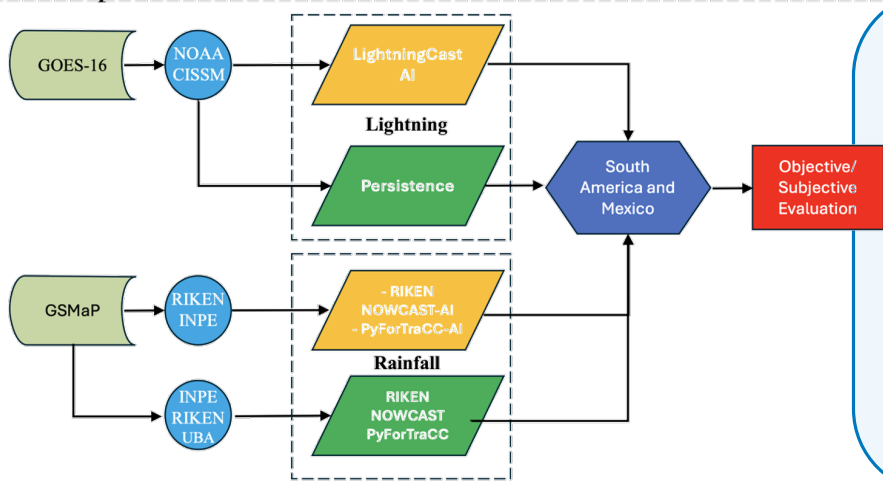
AINPP LATAM

Objectives

Main Objective:

Deploy and evaluate AI-based nowcasting models for Latin America, assessing their effectiveness and operational feasibility.

Intercomparison in Latin America



Specific Objectives:

- Evaluate AI models for lightning and precipitation nowcasting.
- Compare AI products with traditional nowcasting techniques.
- Promote best practices and define standard procedures for AI-based nowcasting.
- Integrate public-private partnerships and enhance regional capacities.

ROADMAP

From regional challenge to operational experimentation

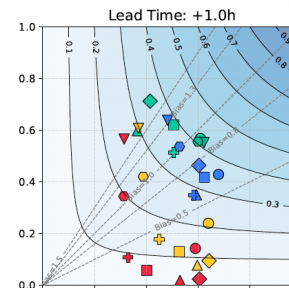
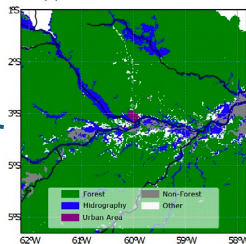
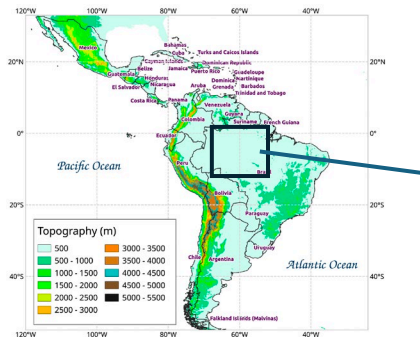
Regional
challenge

Common
benchmark

Lightning &
precipitation

Verification
& lessons

R2O:
PrecSat / NiB



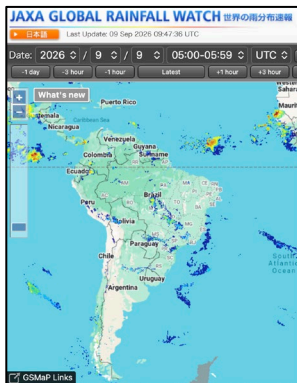
How do we make AI nowcasting useful under heterogeneous observations and operational capacities?

AINPP LATAM DESIGN

One framework, complementary observing regimes

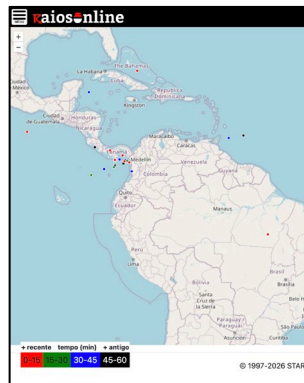
Continental precipitation

- GSMaP benchmark
- 0.1° / hourly
- +1 to +6 h



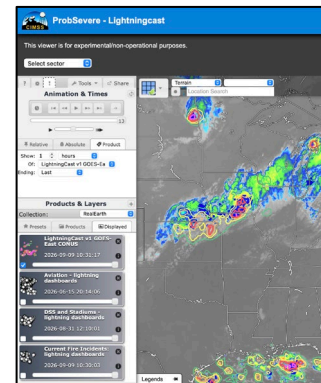
Ground-based lightning

- STARNET + SIPAM radar
- ~2 km / 6–12 min
- +6 and +12 min



Satellite LightningCast

- Satellite perspective
- Mexico + South America
- John Cintineo



Cross-cutting: common verification • meaningful baselines • regional adaptation

FUTURE DEVELOPMENTS

From regional testbed to reusable nowcasting components

Higher-resolution
satellite products

Radar + lightning
products

Common regional
verification

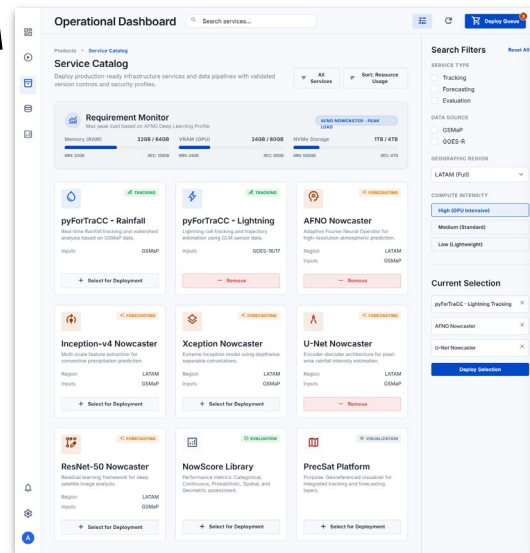
Standardized
containers

AINPP Nowcasting-in-a-Box (NiB)

- Contribute regionally evaluated products, tracking capabilities and verification tools.
- Portability and reproducibility are priorities, but operational scaling also requires orchestration, synchronization and monitoring.

Trying to decrease the uncertainties

- New/Adjusted Satellite Precipitation Estimation Products (QPE) as INPUT for QPF (e.g. IPWG Group)



Development of AI Technical Guidelines

Collaboration with WWRP/WCRP

- Following AI Exploration Roadmap for WIPPS, SC-WIPPS initiated development of AI Technical Guidance, in collaboration with WWRP&WCRP

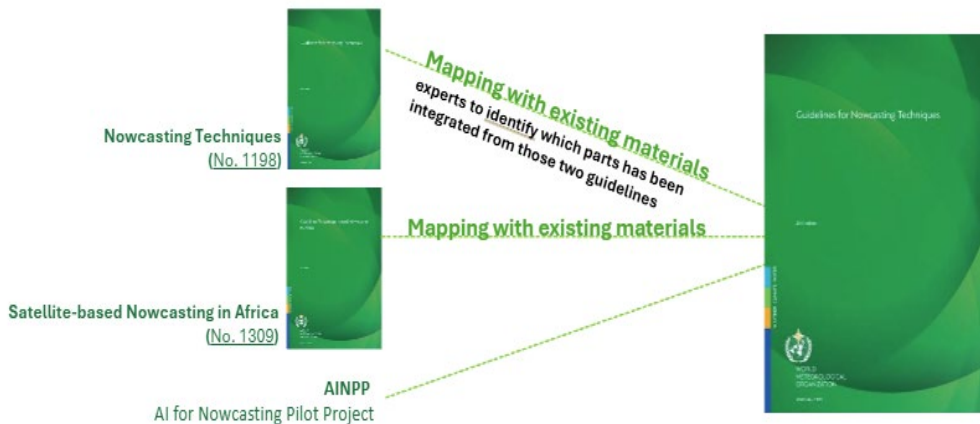
- Nowcasting
- Medium-range forecasts

- Nowcasting

- Update the existing Guidelines on Nowcasting Technique and Satellite-based Nowcasting in Africa.
- Incorporate guidelines on AI-based nowcasting considering the outcomes of AINPP

- Medium-range forecasts

- Developing a new guidance on the use of AI by addressing challenges of AI Exploration Roadmap for WIPPS.
- Using the outcomes of WP-MIP and AI-TC projects.



New WIPPS Strategy – three pillars

Modernize Operational Infrastructure

- Reliable, sustainable and efficient WIPPS infrastructure supporting physics-based, AI-based, and hybrid Earth system prediction workflows

Strengthen Capacity Development to Maximize the Benefit of WIPPS

- Enhanced Members' ability to access, interpret, and utilize WIPPS products, with targeted support for LDCs, SIDS

Advance Innovation and Research to Operation

- Smooth transition of scientific advances especially AI-ESP into sustainable operations with clear governance and pilot programs

Incorporate AI into WIPPS

Value cycle

AI in WIPPS: strategic aspects

- AI-ESP development is rapid and future operational configurations uncertain
- WIPPS must be able to support AI, physics-based and hybrid workflows with appropriate verification and quality assurance
- Transparency, interpretability and trust
- Ensuring equitable access and capacity across Members
- Challenge/benefits of non-traditional sources
 - Provide clear benchmarking guidance for evaluation of products from non-traditional sources
- How much will AI 'democratize' ESP? Will Members still need to rely on cascade of WIPPS products and support?

Sharing the progress of AI developments

- [WMO AI Webinars](#) since 2026
 - Sharing national experience and knowledge on AI application developments
 - 5th Webinar for 13-14UTC on 24 September
 - Good practices identified through ECMWF AI Weather Quest
 - Registration from [here](#)

Issue No. 4 - 2026 (1)

Published in Feb 2026*

*Updated 2 March 2026 (Added Article #12: RSMC Hong Kong for Nowcasting)

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#WIPPS_Pilot_Project #AI #Model_Verification #R20

[03. ECMWF AI Weather Quest: Completion of the First Forecasting Period](#)

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#Weather_Domain #Nowcasting #AI #WIPPS-DC

FIRST WMO AI WEBINAR

22 JANUARY 2026
10:00 - 14:00 UTC

INSEON LEE
(KMA, Republic of Korea)

Skills at long lead forecasting with ensemble AI representations

EMILIA BAGNARDI
(ECMWF)

Integration of AI into operational weather and environmental forecasting systems at ECMWF

REGISTRATION

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SECOND WMO AI WEBINAR

Regional Focus: West Africa

Advancing Nowcasting with Deep Learning Techniques (AIModel) for West Africa
Speaker: Jeffrey H. A. Ayres (RSMC, China)

WAS-NowCast: An Objective Multi-Method Framework for Seasonal Climate Forecasting in West Africa
Between Hydrological Models and Artificial Intelligence (AI) (JCMR/ET-CC-WAF, Niger)

27 April 2026, 8:00 - 9:00 UTC

QR CODE

WMO

AI Weather Quest:
A year of progress in AI-based sub-seasonal prediction

Webinar
24 September 2026 | 1300 UTC

AI Weather Quest ECMWF

Third WMO AI Webinar

16 June 2026
7 - 8 UTC

Earth Sciences
Dev. Zealand

Tristan Meyers
Earth Sciences New Zealand

Operational subseasonal to seasonal (S2S) prediction of drought using ML Downscaling

Wai Kin WONG
Hong Kong Observatory

Integrating AI in Seamless Prediction from Nowcasting to Medium-Range Weather Forecast

REGISTRATION from HERE

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Fourth WMO AI Webinar:
Energy Services in South America Supported by the ENANDES Project

AI for Estimating
Renewable
Energy
Forecasting
Systems (AIRES)

ML-Based
Estimation of Solar
Radiation Forecasts
preparing

AI-Based Downscaling of Climate
Energy Allocations (Climate, Peru, Ecuador, Chile, Argentina)

Thursday, 30 July 2026 | Online
12:00-13:00 UTC
English-Spanish simultaneous interpretation

QR CODE

Authority, trust and accountability



Leveraging AI to enhance MHEWS

- 1. Invest in foundational infrastructure**
Strengthen observation networks, data systems, computing capacity and AI literacy.
- 2. Anchor AI in governance and accountability**
Define focal points, human oversight, validation processes and recourse mechanisms.
- 3. Prioritise human-centred, equity-driven design**
Co-design with affected communities and ensure access in low-connectivity contexts.
- 4. Design AI enabled EWS with interpillar integration from the outset**
Use interoperable architectures, shared data platforms and feedback loops across all pillars.
- 5. Mobilise funding to scale country-led pilots**
Move tested AI solutions into operational workflows through sustained financing and partnerships.



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